

Advanced Algorithms (Quiz 1)

Course Code: CSE426

Total Marks: 20

Section 2

ID: _____ Name: _____ Serial: _____

Questions

- Q1. (8 Marks)** Draw a simple graph with the degree sequence 5,2,2,1,1,1. Show intermediate steps.
- Q2. (9 Marks)** Prove that for any simple graph with 7 vertices G or G' contains an induced C_3
- Q3. (3 Marks)** Let's say you want to color a wheel graph, W_n with 2 colors. For what values of n , it is possible to do that? Give supporting logic to your answer.
- Q4. (3 Marks)** Prove that, complement of a wheel graph is always a disconnected graph.
- Q5. (2 Marks)** Let's say you have 5 vertices a, b, c, d, e . You want to draw a simple undirected graph with these 5 vertices. But there is a constraint that a is allowed to have edges with only c, e . How many possible graphs can be drawn? Show necessary calculations.